

MMF-Net: the key points

A Day Before the Flood: Terrain-Aware Neural Forecasting of River High-Flow Events in Sri Lanka

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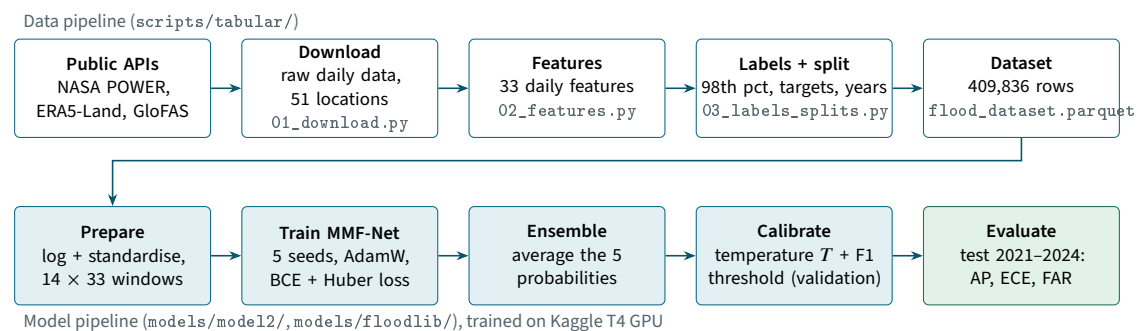
Our work in one sentence. We built a terrain-aware transformer that reads 14 days of weather, soil and river data for a location and predicts the probability that the river will be unusually high **tomorrow**, at 51 locations across Sri Lanka.

Headline numbers. AP **0.8355** (our first model: 0.742; a plain transformer: 0.608) · calibration error **0.0016** · closes 87% of the gap to the LightGBM tree baseline.

1 Task and data

- **Target:** 1 if tomorrow's river discharge is above that location's **98th percentile** (fitted on 2003–2017), else 0. Only about **2% of days are positive**: imbalanced binary classification.
- **Data (free, public, downloaded with our scripts):** NASA POWER (weather, soil wetness), ERA5-Land via Open-Meteo (rainfall), Copernicus GloFAS via Open-Meteo (river discharge). 51 locations, 16 basins, daily 2003–2024: 409,836 location-days, 1,464 high-flow episodes.
- **Input:** last 14 days × 33 daily features (11 rain, 7 weather, 6 soil, 9 discharge) + 9 fixed location facts (elevation, river size, climate zone, river position).
- **Split by time:** train 2003–2017, validation 2018–2020, test 2021–2024. Thresholds and normalisation use training years only.

2 System flow: from raw data to a prediction



3 Research flow: how we reached MMF-Net

Step	What we did	Test AP	What we learned
1. Baselines	Persistence, climatology, discharge-percentile rule, LightGBM, all on the same data	0.590 / 0.108 / 0.816 / 0.850	Today's discharge already predicts tomorrow well. LightGBM sets the bar.
2. Model 1 (TF-STGNN)	GRU over 14 days + graph attention (GATv2) on the river network + FiLM terrain	best 0.742	Lost to trees by 0.108. FiLM helped most (+0.105); the graph helped little.
3. Diagnose	Asked why trees win	–	Raw numbers are a weak input for neural nets (Gorishniy et al., 2022). A GRU passes rain information step by step.
4. Model 2 (MMF-Net)	PLR embeddings + transformer + feature attention + FiLM, no graph; tested one change at a time (N0–N6)	0.8355	Closed 87% of the gap to trees. The input layer was the key. Focal loss was a defect.
5. Check the graph (Model 3)	Added Model 1's river graph to MMF-Net	0.7527	The graph lowers AP, so graph-free is the right choice.

6. Matched re-test	Same test rows, LightGBM vs. MMF-Net (3 seeds)	0.8606 vs. 0.8554	Very close on 1-day; MMF-Net is best on 2-day, 3-day and onset.
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Steps 1–2 used a slightly older data version; steps 5–6 are extra work not in the paper.

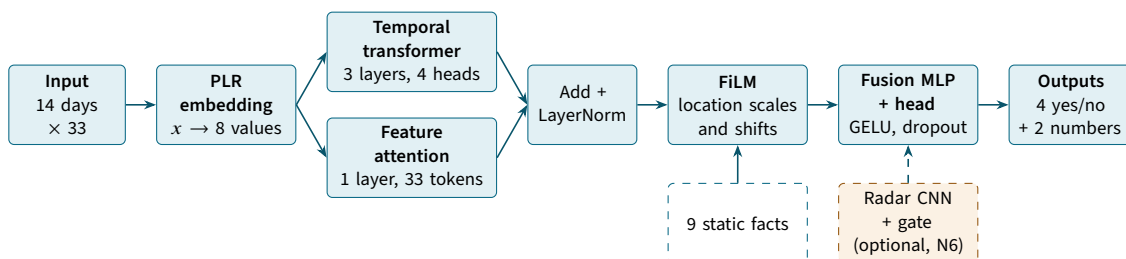
4 Our baselines (what we compare against)

Baseline	How it works	AP, first run	AP, matched
Persistence	Tomorrow's state = today's state	0.590	0.596
Climatology	How often this location is high on this day of the year (training years)	0.108	–
Discharge-percentile rule	Today's discharge percentile used as the score. This is what GloFAS already tells an operator for free.	0.816	0.818
LightGBM (gradient-boosted trees)	600 trees, 63 leaves, class-balanced. Inputs: the 33 features at 0, 1, 3, 7, 14 days back + 9 static = 174 numbers	0.850	0.861
N0, DNN starting point	Our transformer with plain linear inputs, no feature attention, no FiLM	0.608	–
Model 1, previous model	GRU + river graph + FiLM, 5-seed ensemble (M5)	0.742	–
MMF-Net	Final model	0.8355	0.855

“Matched”: a later run (September 2026, not in the paper) with identical test rows for every method; MMF-Net used 3 seeds and a different validation/calibration split, so its score differs from the paper's 0.8355.

BE CAREFUL The paper itself does not rank MMF-Net against LightGBM, because the first-run baselines used an older data version. If asked, quote the matched run and say it is not in the paper.

5 Architecture, and why we chose each part



One shared network for all 51 locations, **about 711,000 parameters**.

Part	What it does	Why we chose it	Effect
PLR embedding	Turns each number into 8 learned sin/cos-based values	Neural nets on raw numbers lose to trees (Model 1: 0.742 vs 0.850). Embeddings are the published fix (Gorishniy et al., 2022).	+0.152 AP , the biggest gain
Temporal transformer	One 128-number token per day + CLS token; self-attention across the 14 days	Model 1's GRU had to carry rain information through 14 steps. Attention looks straight back at the rain day.	About equal to the GRU alone (0.608 vs 0.598)
Cross-feature attention	The 33 features attend to each other	One linear layer cannot express “rain matters only when the soil is already wet”.	+0.013 AP
FiLM terrain conditioning	9 location facts give a scale γ and shift β for each of the 128 values	The same rain gives a different river response at each location. FiLM was Model 1's best step (+0.105).	+0.053 AP ; false alarms 0.313 $\rightarrow$ 0.110
No graph	Each location is processed separately with shared weights	Model 1's graph gave little; we wanted a strong single-location model first.	Model 3: adding the graph gave -0.083 AP
Gated radar (optional)	Sentinel-1 image $\rightarrow$ CNN $\rightarrow$ added through a learned gate	Model 1 concatenated images and gained nothing; about 92% of days have no fresh image.	No gain (0.8310 vs 0.8355), 17 $\times$ more parameters

$$\text{PLR: } e_j(x) = \text{ReLU}(W_j [\sin(2\pi c_j x) \parallel \cos(2\pi c_j x)] + b_j) \quad \text{FiLM: } \tilde{h} = \text{LayerNorm}(h \odot (1 + \gamma) + \beta)$$

6 Loss function and training

Six outputs are learned together (multi-task): high flow tomorrow (**the main one, the only one scored**), within 2 days, within 3 days, a new event starting tomorrow (onset), and 2 regression values (next-day discharge percentile, 3-day max discharge z-score).

$$\mathcal{L} = 1.0 \text{ BCE}_{1\text{-day}} + 0.3 \text{ BCE}_{2\text{-day}} + 0.3 \text{ BCE}_{3\text{-day}} + 0.5 \text{ BCE}_{\text{onset}} + 0.2 \text{ Huber}$$

$$\text{BCE} = -[y \log p + (1 - y) \log(1 - p)] \quad \text{Huber}(e) = \begin{cases} \frac{1}{2} e^2 & |e| \leq 1 \\ |e| - \frac{1}{2} & \text{otherwise} \end{cases}$$

Why BCE and not focal loss? We first used focal loss, $\text{FL} = -\alpha_t (1 - p_t)^\gamma \log p_t$ ($\alpha = 0.75$, $\gamma = 2$, plus a label-confidence weight), because it is the usual choice for rare classes. It made results worse: AP **0.8269** $\rightarrow$ **0.7592** and calibration error **0.0032** $\rightarrow$ **0.0525**. Switching back to BCE gave our best model.

Training: PyTorch; AdamW (lr 10^{-3} , weight decay 10^{-4}); 5 warm-up epochs then cosine decay; max 60 epochs; early stopping (patience 10) on validation; gradient clipping 1.0; batch 32 days $\times$ 51 locations. **Final model:** 5 seeds averaged, then temperature scaling $p_{\text{cal}} = \sigma(\text{logit}(\bar{p})/T)$; alarm threshold = best validation F1.

7 What we did to improve, step by step (test 2021–2024)

ID	What we changed	AP $\uparrow$	Change	ECE $\downarrow$	FAR $\downarrow$
N0	Start: transformer with plain linear inputs	0.6083	–	0.0024	0.412
N1	Replace plain inputs with PLR embeddings	0.7605	+0.152	0.0053	0.325
N2	Add cross-feature attention	0.7738	+0.013	0.0056	0.313
N3	Add FiLM terrain conditioning	0.8269	+0.053	0.0032	0.110
N4	Try focal loss instead of BCE	0.7592	–0.068	0.0525	0.138
N5	N4 + 5-seed ensemble + temperature scaling	0.7846	–	0.0043	0.145
N5_bce	Back to BCE: N3 + 5-seed ensemble + temperature scaling	0.8355	+0.009	0.0016	0.120
N6	N5 + radar CNN with gate	0.8310	–	0.0066	0.107

Change = difference from the row above (N5_bce is compared with N3). N0–N4: 1 seed; N5, N5_bce, N6: 5 seeds. AP: average precision (random guess $\approx$ 0.02). ECE: calibration error. FAR: share of alarms that were false.

Total improvement:

- From the plain transformer (N0) to the final model: AP **0.608** $\rightarrow$ **0.836**, false alarms **0.412** $\rightarrow$ **0.120**.
- From our first model (Model 1) to MMF-Net: AP **0.742** $\rightarrow$ **0.836**; the gap to LightGBM shrank from 0.108 to 0.014 (**87% closed**). Model 1 used slightly older data and focal loss, so this is not a perfectly controlled comparison.
- Calibration: ECE **0.0016**, Brier 0.0080, the best of all our models.

8 Comparison with other research

Study	Scale and method	Reported	Important detail
Saubhagya et al. 2025	Ratnapura (Kalu), 1 station + 4 gauges, daily 2014–2019; Bi-LSTM + regression, predicts water level	80%, 80%, 100% accuracy (1–3 days)	Only 5 flood events in the test years; good time-based split
Thilakarathne & Premachandra 2017	Anuradhapura district, monthly 1976–2015; ARIMA + small ANN	91.7% accuracy	Mean recall 0.404; random 10-fold split
Mahaganapathy et al. 2023	6 Kalu catchments; SVM, trees, bagging, SMOTE	99% F1	Abstract only; split not stated
This work	51 locations, 16 basins, daily 2003–2024; transformer + PLR + FiLM	AP 0.8355, ECE 0.0016	Time-based split; 1,464 episodes

These numbers **cannot be compared directly** (different targets, places and metrics). Our difference: national scale, a leak-free time split, and AP plus calibration instead of accuracy (with 98% “no” days, always saying “no” scores 98% accuracy). State their numbers as facts, not criticism; only Saubhagya et al. is cited in our paper.

International work: Kratzert et al. 2018 showed an LSTM can learn river response from data; we use a transformer. Nearing et al. 2024 (Google, *Nature*) is global and operational; ours is national with public data only. Gorishniy et al. 2022 found number embeddings help neural nets on tables; **our result agrees** (PLR +0.152). Kirschstein & Sun 2024 found river graphs did not help; **our Model 3 agrees**.

9 What we do best

1. **Trustworthy probabilities.** ECE 0.0016 and Brier 0.0080: on average, predicted probabilities match how often high flow really happens. None of the Sri Lankan studies we reviewed report calibration.
2. **A better input layer.** PLR embeddings gave the biggest single gain and brought the network close to tree models.
3. **One model for the whole country.** FiLM adapts a single network to each location and cut false alarms by about two thirds.
4. **Scale with honest evaluation.** 51 locations, 16 basins, 22 years, split by time. Negative results (focal loss, radar, graph) are reported, not hidden.
5. **Small and efficient.** 0.71 M parameters; the radar model is 17× bigger with no gain.

10 Limitations (say these before she does)

- The target is **GloFAS reanalysis discharge**, not measured water levels or flood maps: we predict high river flow, a proxy for flood hazard.
- N0–N4 are **single runs** with no confidence intervals. Focal loss changed three things at once; ensemble and temperature scaling were added together.
- Radar covers only 9 of 51 locations (an image about every 12 days) and used focal loss, so it is not a fair test.
- Not yet an operational warning system: real-time data delays were not simulated.

11 Likely questions

Why this architecture? Our first model lost to LightGBM (0.742 vs 0.850). We found the cause was the input layer, so we added PLR embeddings, and kept FiLM because it was the strongest part of Model 1. Every part was tested by adding it alone (section 7).

What is your baseline? Four standard methods (persistence, climatology, discharge-percentile rule, LightGBM), our plain-transformer start N0, and our previous Model 1 (section 4).

Why split by time, not randomly? Neighbouring days are nearly identical, so a random split leaks. In Model 1, a random split raised AP from 0.704 to 0.905 while early-warning detection fell from 0.380 to 0.089.

Why not predict water level and then threshold, like Saubhagya et al.? There are no public gauge data at 51 locations. We predict a calibrated probability directly and measure it with AP, which shows performance on rare events honestly.

Why the 98th percentile? About 7 days a year per location: rare, but enough examples to learn from. Per location, so small and big rivers are treated fairly. Not an official warning level.